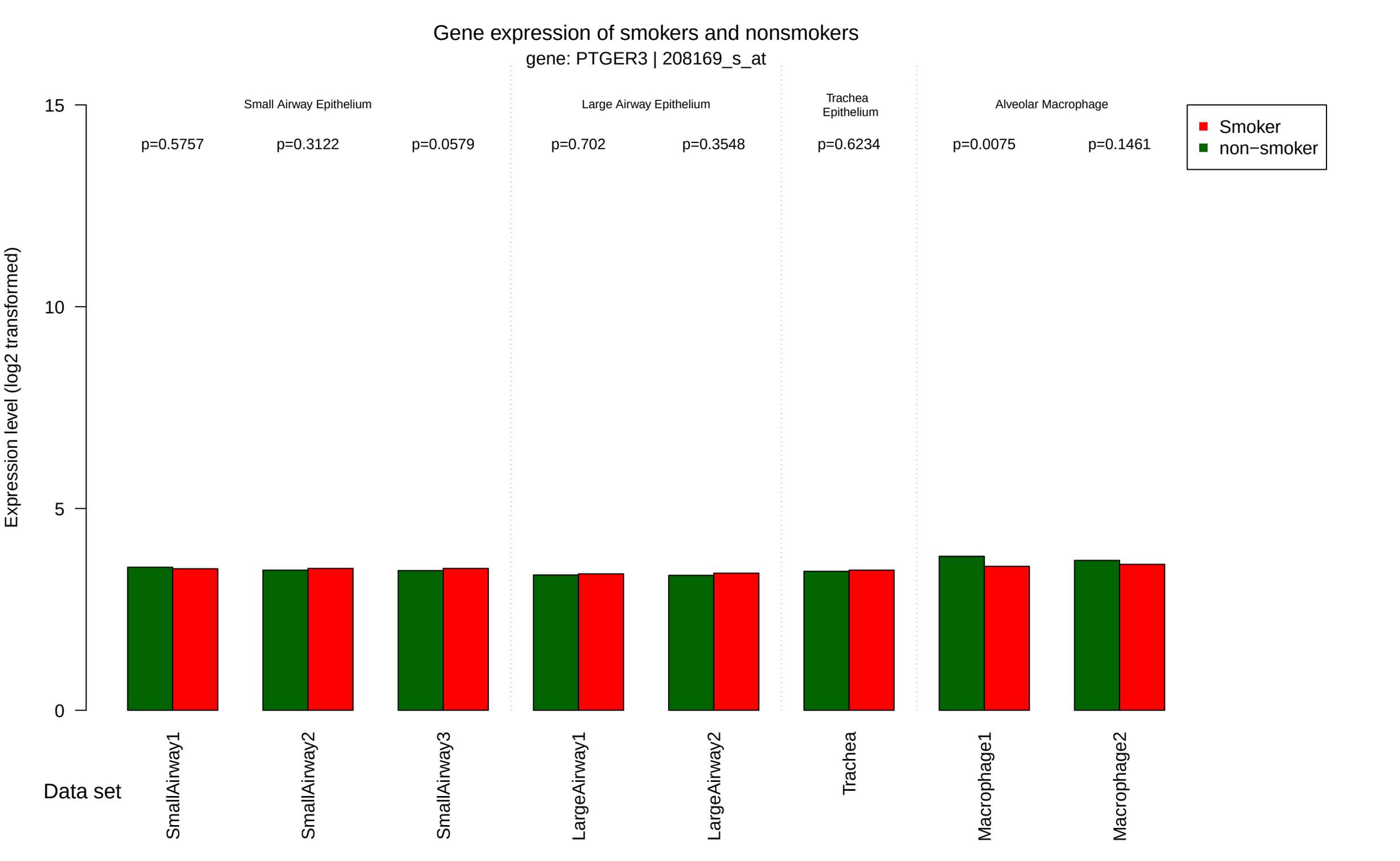
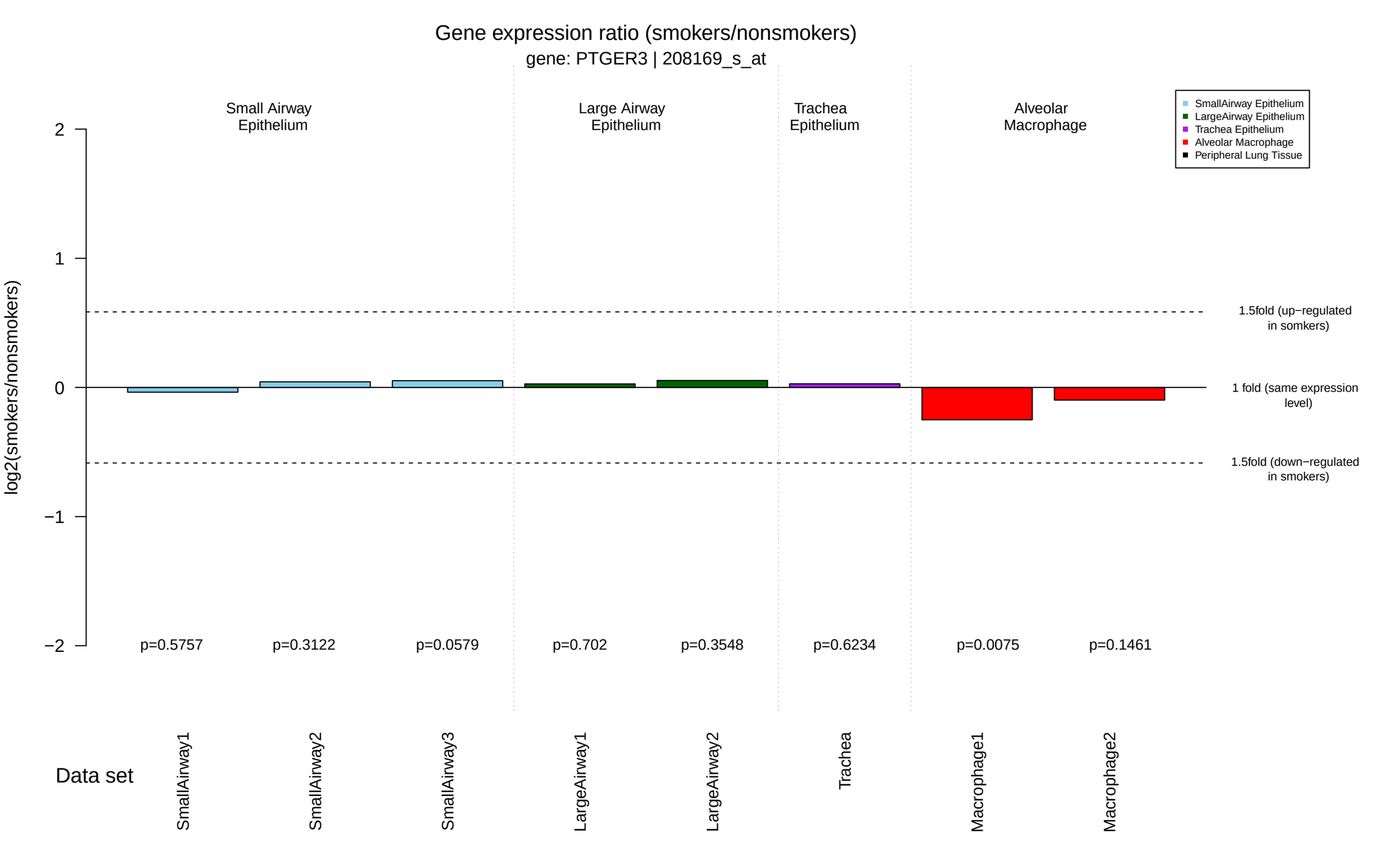
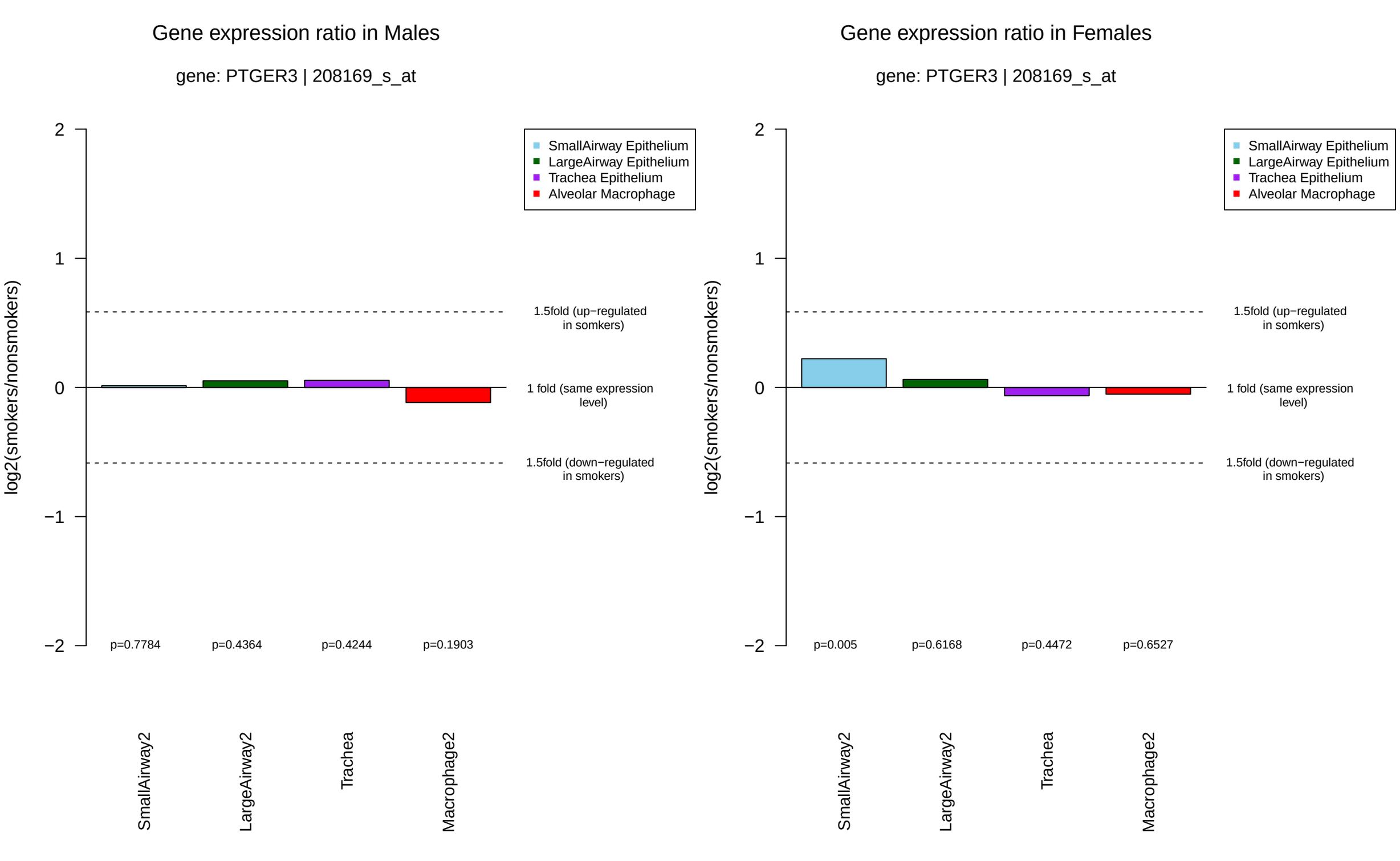
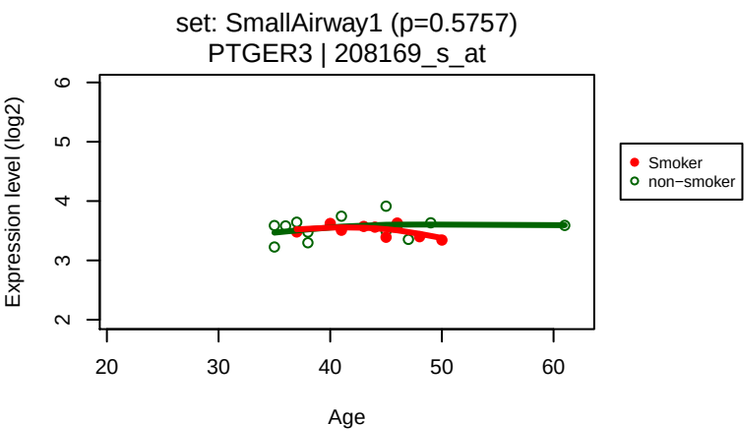
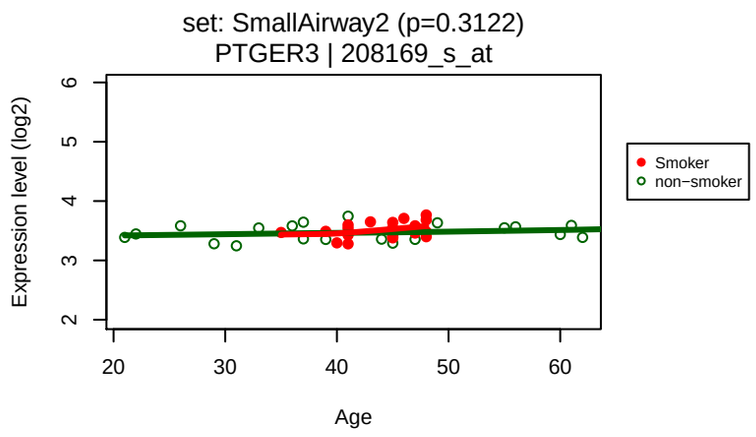
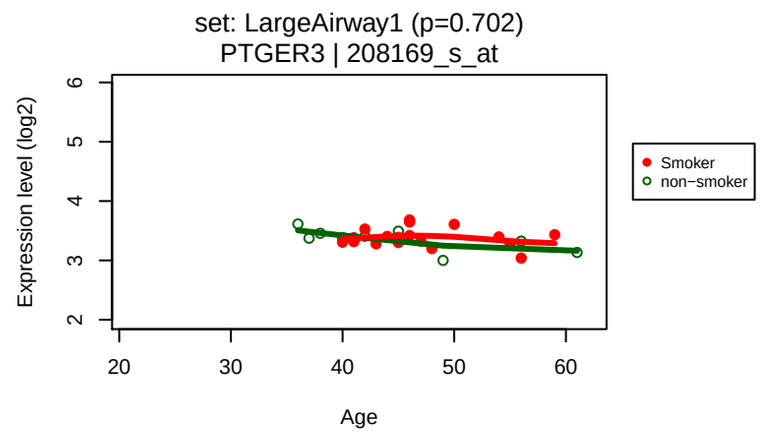

Gene expression of smokers and nonsmokers
gene: PTGER3 | 208169_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5757
p=0.3122
p=0.0579
p=0.702
p=0.3548
p=0.6234
p=0.0075
p=0.1461
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

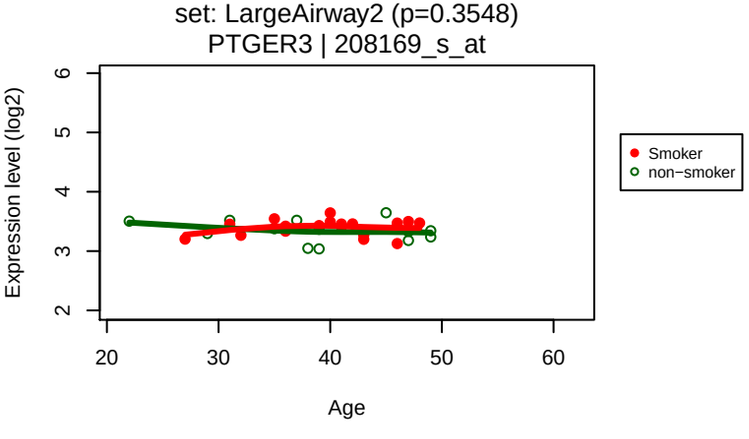

Gene expression ratio (smokers/nonsmokers)
gene: PTGER3 | 208169_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5757
p=0.3122
p=0.0579
p=0.702
p=0.3548
p=0.6234
p=0.0075
p=0.1461
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

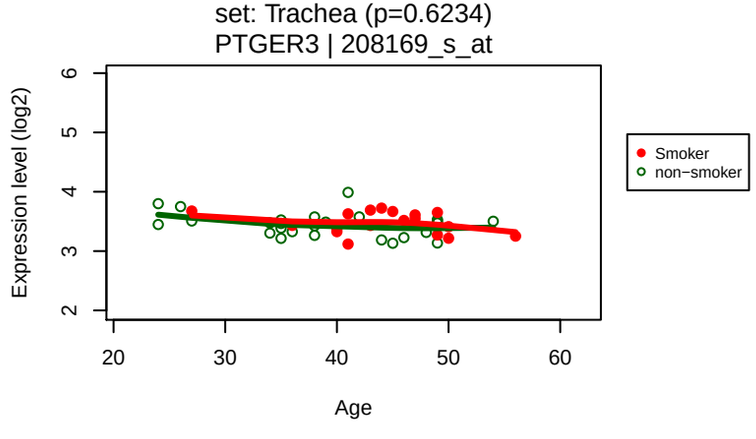

Gene expression ratio in Males
gene: PTGER3 | 208169_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7784
p=0.4364
p=0.4244
p=0.1903
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTGER3 | 208169_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.005
p=0.6168
p=0.4472
p=0.6527
SmallAirway2
LargeAirway2
Trachea
Macrophage2

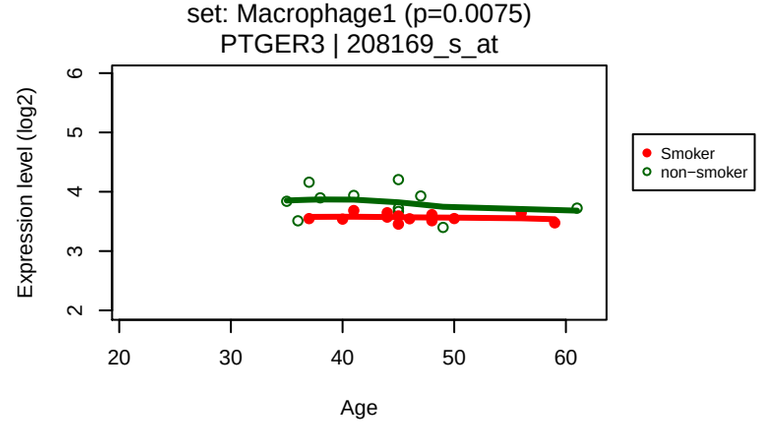

set: SmallAirway1 (p=0.5757)
PTGER3 | 208169_s_at
Expression level (log2)
Age
Smoker
non−smoker

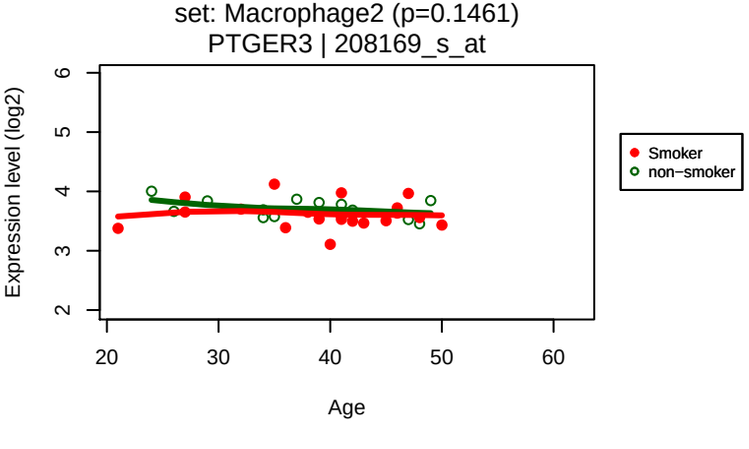

set: SmallAirway2 (p=0.3122)
PTGER3 | 208169_s_at
Expression level (log2)
Age
Smoker
non−smoker

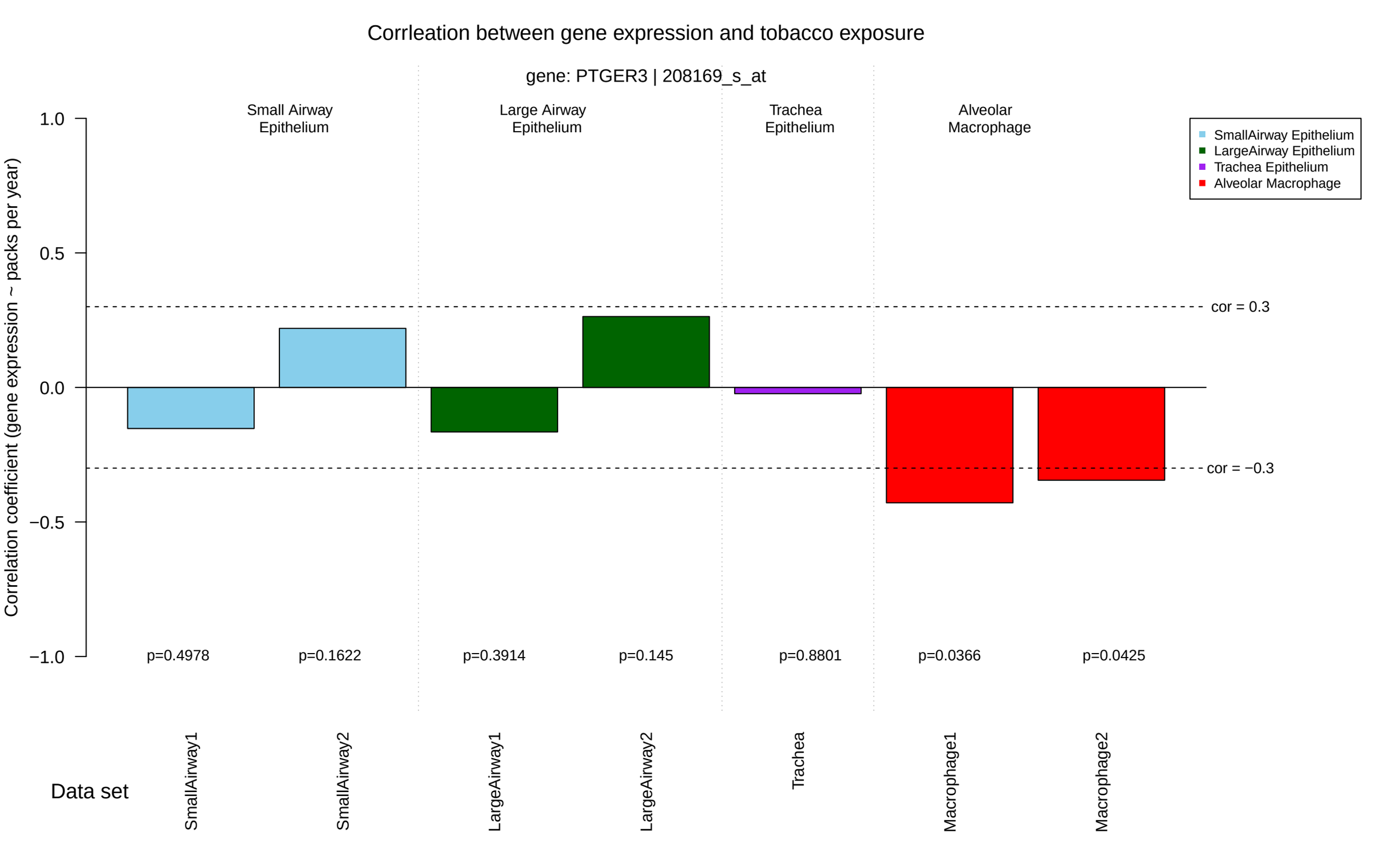

set: LargeAirway1 (p=0.702)
PTGER3 | 208169_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3548)
PTGER3 | 208169_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6234)
PTGER3 | 208169_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0075)
PTGER3 | 208169_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1461)
PTGER3 | 208169_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTGER3 | 208169_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4978
p=0.1622
p=0.3914
p=0.145
p=0.8801
p=0.0366
p=0.0425
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2